

PRACTICAL WEEK 1

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- Write a C code to create a process using fork() system call.
- Read the user input as shell command.
- Use a system call exec() to execute the shell command.
- Use a wait() system call for waiting a parent process.
- Display the PIDs of parent and child processes.

CODE:

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/wait.h>
6 #include <string.h>
7
8 int main()
9 {
10     char command[100];
11     pid_t pid;
12
13     printf("Enter a Linux command: ");
14     fgets(command, sizeof(command), stdin);
15
16     // Remove newline character
17     command[strcspn(command, "\n")] = '\0';
18
19     // Create child process
20     pid = fork();
21
22     if (pid < 0)
23     {
24         perror("fork failed");
25         return 1;
26     }
27     else if (pid == 0)
28     {
29         // Child process
30         printf("\nChild Process\n");
31         printf("Child PID: %d\n", getpid());
32         printf("Parent PID: %d\n", getppid());
33
34         // Execute command
35         exec("/bin/sh", "sh", "-c", command, (char *)NULL);
36
37         // Only runs if exec fails
38         perror("exec failed");
39         exit(1);
40     }
41     else
42     {
43         // Parent process
44         printf("\nParent Process\n");
45         printf("Parent PID: %d\n", getpid());
46         printf("Child PID: %d\n", pid);
47
48         // Wait for child
49         wait(NULL);
50
51         printf("\nChild process completed.\n");
52     }
53
54     return 0;
55 }
```

OUTPUT:

```
Last login: Sat Aug 25 18:00:51 on ttys000
screenkhilreddy@MacBookPro ~ % nano 100.c
screenkhilreddy@MacBookPro ~ % gcc 100.c -o 100
screenkhilreddy@MacBookPro ~ % ./100
Enter a Linux command: pwd

Parent Process
Parent PID: 1137
Child PID: 1145

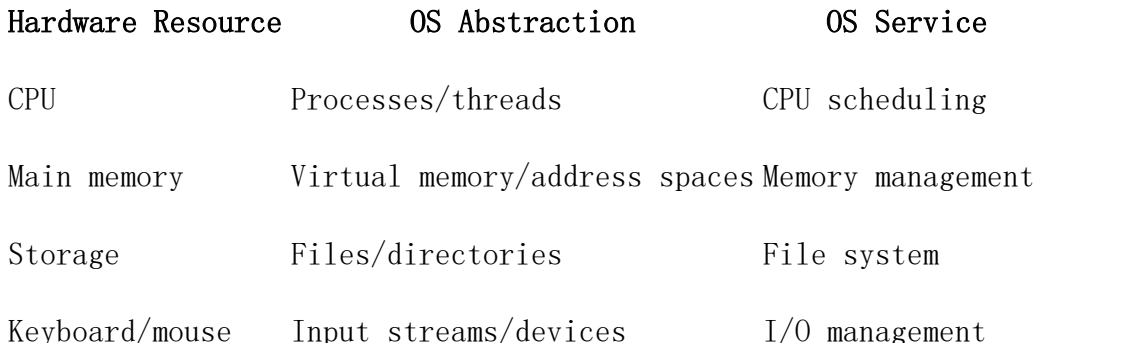
Child Process
Child PID: 1145
Parent PID: 1137
Parent PID: 1137
/Users/screenkhilreddy

Child process completed.
screenkhilreddy@MacBookPro ~ % █
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
screenkhilreddy@MacBookPro ~ % ps
PID TTY          TIME CMD
1114 ttys000    0:00.04 -zsh
screenkhilreddy@MacBookPro ~ % █
```

```
screenkhilreddy@MacBookPro ~ % ps
PID TTY          TIME CMD
1111 tty000      0:00.04 -zsh
screenkhilreddy@MacBookPro ~ % ps aux
USER          PID  CPU   MEM      VSZ    RSS  TT  STAT  STARTED      TIME COMMAND
screenkhilreddy  808  0.0   0.1 36143156 808652  77  S   10:53AM  2:09.91 /Applications/psoffice.app/Contents/MacOS/psoffice
screenkhilreddy  805  3.3  1.0 30843492 407128  71  S   10:53AM  0:21.59 /Applications/psoffice.app/Contents/SharedSupport/pscloudsvr.app/Contents/MacOS/pscloudsvr Run /InstanceId=WpsCloudSvr wpscloudsvrplugin.dll /qingbangong
root          145  3.2  0.1 33839444 16288  71  S   10:52AM  0:11.88 /usr/libexec/dand
root          116  0.2  0.2 33978548 26212  71  S   10:52AM  0:11.84 /usr/libexec/TouchBarButton
screenkhilreddy 1178  3.1  0.2 33980148 31268  71  R   10:53AM  0:16.27 /usr/libexec/dutepertid
root          147  2.0  0.0 34613992 63288  71  S   10:52AM  0:08.45 /System/Library/PrivateFrameworks/SkyLight.framework/Resources/WindowServer -daemon
screenkhilreddy 409  1.0  0.3 34658784 54632  77  S   10:52AM  0:06.27 /System/Library/CoreServices/NotificationCenter.app/Contents/MacOS/NotificationCenter
root          179  1.2  0.1 33845492 15564  71  S   10:52AM  0:02.62 /System/Library/PrivateFrameworks/ContextMenu.framework/Resources/ContextMenu
screenkhilreddy 977  1.0  0.7 34759840 121348  77  S   10:56AM  0:08.07 /System/Applications/Utilities/Terminal.app/Contents/MacOS/Terminal
root          152  0.5  0.1 33845492 15564  71  S   10:52AM  0:09.04 /usr/libexec/openssh/ssh-keygen
root          148  0.5  0.1 33868888 21748  71  S   10:52AM  0:09.38 /usr/libexec/locationd
screenkhilreddy 837  0.5  1.0 88202228 272744  77  S   10:52AM  0:28.48 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/151.0.7922.138/Helpers/Google Chrome Helper.app/Contents/MacOS/Google Chrome Helper
root          193  0.4  0.3 51599480 127544  71  S   10:52AM  0:19.83 /System/Library/Frameworks/CoreServices.framework/Frameworks/Metadata.framework/Versions/A/Support/mds_stores
screenkhilreddy 1163  0.3  0.0 33795572 7720  77  S   10:56AM  0:04.49 /System/Library/Frameworks/CoreServices.framework/Frameworks/Metadata.framework/Versions/A/Support/mdworker_shared -s mdworker -c MDISImporterWorker -m com.apple
screenkhilreddy 254  0.3  2.0 34732968 407392  77  S   10:52AM  0:00.51 /usr/local/mysql/bin/mysql --user=mysql --basedir=/usr/local/mysql --datadir=/usr/local/mysql/data --plugin-dir=/usr/local/mysql/plugin --log-error=/usr
screenkhilreddy 966  0.2  3.0 154786874 582654  77  S   10:55AM  0:50.71 /Applications/Google Chrome.app/Contents/Frameworks/Google Chrome Framework.framework/Versions/151.0.7922.138/Helpers/Google Chrome Helper (Renderer).app/Con
root          121  0.1  0.3 34848776 47788  77  S   10:52AM  0:13.12 /System/Library/Frameworks/CoreServices.framework/Frameworks/Metadata.framework/Support/mds
screenkhilreddy 441  0.1  0.9 34737322 151756  77  S   10:52AM  0:20.92 /System/Library/CoreServices/Finder.app/Contents/MacOS/Finder
screenkhilreddy 915  0.1  1.9 155479112 514088  77  S   10:53AM  0:28.78 /Applications/Google Chrome.app/Contents/MacOS/Google Chrome
root          288  0.1  0.0 34217468 5944  71  S   10:52AM  0:00.79 /usr/libexec/audiolockdown
root          1386  0.0  0.0 33816272 7176  71  S   10:53AM  0:06.87 /System/Library/PrivateFrameworks/XProtect.framework.Framework/Versions/A/XPCServices/XProtectUpdateService.xpc/Contents/MacOS/XProtectUpdateService
root          1305  0.0  0.0 33773224 3532  71  S   10:55AM  0:00.03 /usr/libexec/trd
root          1217  0.0  0.0 33738824 3508  71  S   10:55AM  0:00.07 /usr/libexec/betacorellentid
screenkhilreddy 1195  0.0  0.1 34268896 11816  77  S   10:53AM  0:00.13 /System/Library/PrivateFrameworks/AppPredictionFoundation.framework/Versions/A/XPCServices/AppPredictionIntentHelperService.xpc/Contents/MacOS/AppPredictionI
screenkhilreddy 1105  0.0  0.0 33818688 3508  77  S   10:53AM  0:00.07 /usr/libexec/betacorellentid
screenkhilreddy 1194  0.0  0.1 33816692 11776  71  S   10:53AM  0:00.19 /System/Library/PrivateFrameworks/IntelligencePlatformCore.framework/Versions/A/IntelligencePlatformCore
screenkhilreddy 1198  0.0  0.1 33857276 11980  77  S   10:53AM  0:00.34 /usr/libexec/prnactived
screenkhilreddy 1189  0.0  0.1 33818824 13812  71  S   10:53AM  0:00.13 /System/Library/PrivateFrameworks/People.framework/people
root          1186  0.0  0.1 33808120 8400  77  S   10:53AM  0:00.21 /usr/libexec/ASPCarryOn
screenkhilreddy 1208  0.0  0.1 34261228 8864  77  S   10:53AM  0:00.04 /System/Library/PrivateFrameworks/CloudElementry.framework/Versions/A/XPCServices/CloudElementryService.xpc/Contents/MacOS/CloudElementryService
screenkhilreddy 1177  0.0  0.1 34089148 13184  77  S   10:53AM  0:00.21 /System/Cryptexes/App/System/Library/CoreServices/SafariSupport.bundle/Contents/MacOS/SafariBookmarksSyncAgent
root          1176  0.0  0.0 33743140 2340  71  S   10:53AM  0:00.02 /System/Library/CoreServices/LiquidDetection.framework/Support/amsdetectionstored
screenkhilreddy 1173  0.0  0.1 33924192 13316  77  S   10:53AM  0:00.20 /System/Library/PrivateFrameworks/OnDeviceStorage.framework/Support/amsdetectionstored
screenkhilreddy 1172  0.0  0.1 33814802 12784  71  S   10:53AM  0:00.24 /System/Library/Frameworks/Translation.framework/translation
screenkhilreddy 1171  0.0  0.1 34288148 12292  77  S   10:53AM  0:00.08 /System/Library/Frameworks/QuickLookUI.framework/Versions/A/XPCServices/QuickLookUIService.xpc/Contents/MacOS/QuickLookUIService
screenkhilreddy 1170  0.0  0.0 34615992 50716  77  S   10:53AM  0:01.04 /System/Library/PrivateFrameworks/AppKit.framework/Versions/O/XPCServices/com.apple.appkit.xpc.openAndSavePanelService.xpc/Contents/MacOS/com.apple.appkit.xpc.openAnd
screenkhilreddy 1168  0.0  0.0 34159468 3432  77  S   10:53AM  0:00.02 /System/Library/Frameworks/ImageCaptureCore.framework/Versions/A/XPCServices/ImageCaptureCore.xpc/Contents/MacOS/ImageCaptureCore
screenkhilreddy 1169  0.0  0.0 33737152 3884  77  S   10:53AM  0:00.04 /usr/libexec/ptcamerad
screenkhilreddy 1166  0.0  0.2 33733240 2232  71  S   10:53AM  0:00.03 /System/Library/Frameworks/NetFS.framework/Versions/A/XPCServices/PlugInLibraryService.xpc/Contents/MacOS/PlugInLibraryService
screenkhilreddy 1165  0.0  0.0 34828212 138356  77  S   10:53AM  0:04.76 /System/Applications/Preview.app/Contents/MacOS/Preview
screenkhilreddy 1164  0.0  0.2 34486712 164904  77  S   10:53AM  0:00.10 /Applications/psoffice.app/Contents/SharedSupport/promacefpuginhost (Renderer).app/Contents/MacOS/promacefpuginhost (Renderer) --type=renderer --framework-
screenkhilreddy 1148  0.0  0.1 33813692 10736  77  S   10:56AM  0:00.32 /System/Library/Extensions/Kit/Extensions/ImageThumbnailExtension.appex/Contents/MacOS/ImageThumbnailExtension
root          1147  0.0  0.0 33810772 2148  77  S   10:56AM  0:00.01 /System/Library/CoreServices/ReportCrash daemon
root          1139  0.0  0.1 33778316 9892  77  S   10:56AM  0:00.78 /System/Library/PrivateFrameworks/XProtect.framework.Framework/Versions/A/XPCServices/XProtectService.xpc/Contents/MacOS/XProtectService
screenkhilreddy 1138  0.0  0.0 33715776 2136  77  S   10:56AM  0:00.01 /usr/libexec/taskgated
root          1114  0.0  0.0 34148548 2352 4880  S   10:56AM  0:00.04 -zsh
root          1113  0.0  0.0 34158924 3388 4880  S   10:56AM  0:00.02 login -pf screenkhilreddy
screenkhilreddy 1097  0.0  0.0 32743210 4444  77  S   10:56AM  0:00.04 /System/Library/Frameworks/CoreSpotlight.framework/Spotlight/MediaIndex
screenkhilreddy 1095  0.0  0.0 33830788 7380  77  S   10:56AM  0:00.07 /System/Library/CoreServices/EscrowSecurityAlert.app/Contents/MacOS/EscrowSecurityAlert
screenkhilreddy 1060  0.0  0.3 33868336 15120  77  S   10:56AM  0:00.38 /System/Cryptexes/App/usr/libexec/AuthenticationServicesAgent
screenkhilreddy 1091  0.0  0.0 33868688 6344  77  S   10:56AM  0:00.08 /System/Library/PrivateFrameworks/Cascode.framework/Versions/A/XPCServices/SetStoreUpdateService.xpc/Contents/MacOS/SetStoreUpdateService
screenkhilreddy 1083  0.0  0.3 154788084 54888  77  S   10:57AM  0:00.13 /Applications/Microsoft Edge.app/Contents/Frameworks/Microsoft Edge Framework.framework/Versions/151.0.4129.78/Helpers/Microsoft Edge Helper (Renderer).app/C
screenkhilreddy 8 1005 0.0 0.9 154787788 151712 77 S 10:57AM 0:01.27 /Applications/Microsoft Edge.app/Contents/Frameworks/Microsoft Edge Framework.framework/Versions/151.0.4129.78/Helpers/Microsoft Edge Helper (Renderer).app/C
screenkhilreddy 0
screenkhilreddy 1078 0.0 0.0 33745528 6880 77 S 10:57AM 0:00.04 /usr/libexec/feedbackd
screenkhilreddy 1076 0.0 0.0 33817972 12424 77 S 10:57AM 0:00.16 /System/Library/PrivateFrameworks/CoreParsec.framework/parsec-daemon
screenkhilreddy 1067 0.0 0.0 33810548 7184 71 S 10:57AM 0:00.13 /System/Library/Frameworks/ExtensionFoundation.framework/Versions/A/XPCServices/extensionkititservice.xpc/Contents/MacOS/extensionkititservice
screenkhilreddy 1066 0.0 0.0 33836248 8368 77 S 10:57AM 0:00.13 /System/Cryptexes/App/usr/libexec/PasswordReachAgent
screenkhilreddy 1065 0.0 0.0 33844888 12480 77 S 10:57AM 0:00.22 /usr/libexec/nlmeta
screenkhilreddy 1064 0.0 0.1 33833152 10444 77 S 10:57AM 0:00.11 /System/Library/PrivateFrameworks/GeoAnalytics.framework/GeoAnalytics
root          1059 0.0 0.0 33735748 2244 77 S 10:57AM 0:00.01 /System/Library/PrivateFrameworks/IntelliDiagnostics.framework/Versions/A/Resources/IntelliDiasd
screenkhilreddy 1044 0.0 0.1 33833216 13892 77 S 10:57AM 0:00.13 /System/IOSupport/System/Library/PrivateFrameworks/HomeEnergyDaemon.framework/Support/homeenergyd
screenkhilreddy 1040 0.0 0.1 33861324 19932 77 S 10:56AM 0:00.70 /usr/libexec/tipsd
screenkhilreddy 1044 0.0 0.1 33812688 19472 77 S 10:56AM 0:00.14 /usr/libexec/diagnosticextensionsd
root          1033 0.0 0.0 34183716 7156 77 S 10:56AM 0:00.06 /System/Library/PrivateFrameworks/RemoteManagement.framework/XPCServices/ManagedConfigurationFilesSubscriber.xpc/Contents/MacOS/ManagedConfigurationFilesSubs
```

[illegible]

Hardware Resource	OS Abstraction	OS Service
Display	Device interface	Device management
Network adapter	Network interface	Network management

